The Attribution Impossibility:

No Importance Ranking Is Faithful, Stable, and Complete Under Symmetry

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Abstract

Attribution instability is not a bug in SHAP—it is a mathematical property of attribution itself, at every level of a model. We prove that no attribution ranking can simultaneously be faithful, stable, and complete under the Rashomon property (the Attribution Impossibility); for binary questions, faithful + stable alone is impossible (the bilemma). The theorem applies to *any* attribution system: input-level (which feature matters) and component-level (which circuit computes). **Input-level:** 50 retrains produce 24 distinct top-3 rankings (4.2% agreement); 45% of German Credit applicants receive different explanation categories; 68% of 77 datasets show instability. A 7-line nonparametric diagnostic outperforms the Gaussian formula by $2\times$. **Component-level:** 10 independently trained transformers on TinyStories agree at only ρ=0.54 on circuit importance by activation patching. A G-invariant projection (orbit averaging over the architectural symmetry group) lifts agreement to ρ=0.98, with 14/14 theory-derived predictions confirmed across two scales. A fine-tuned GPT-2 confirms the boundary: without Rashomon diversity, instability vanishes (ρ=0.993). The design space has exactly two achievable families; DASH (orbit averaging) is Pareto-optimal. Verified in Lean 4 (357 theorems, 6 axioms, 0 `sorry`).

1 Introduction

Train a gradient-boosted model. Compute SHAP values. Retrain with a different random seed. The "most important feature" changes—in 68% of 77 public datasets. Now look inside a neural network. Identify the most important circuit component by activation patching. Retrain from a different initialization. The most important component changes too—10 independently trained transformers on TinyStories agree at only $\rho = 0.54$.

The instability is not in SHAP, not in activation patching, and not in any specific method. It is in the mathematics of attribution itself. We prove that *no* attribution ranking—at any level of a model, from input features to internal circuits—can simultaneously be faithful, stable, and complete when the Rashomon property holds. The same theorem, the same resolution, the same diagnostic, at different granularities.

Two levels, one impossibility. Consider *input-level attribution*: across 30 retrains, 45% of German Credit applicants receive a different "most important feature" category. The adverse action notice changes based solely on the training seed. Now consider *component-level attribution*: across 10 independently trained transformers, all 4 attention heads in layer 0 appear as "most important" under different seeds—full S_4 symmetry realized. A circuit analysis depends on which equivalent network you happened to train.

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The reason is the same: collinear features (or architecturally symmetric components) create a Rashomon set where equivalent models attribute importance in opposite orders. The fix is the same: *orbit averaging*—aggregate across the symmetry to recover what is shared. For features, this is DASH (ensemble averaging); for circuits, this is the G-invariant projection. Both are the Reynolds operator on different symmetry groups.

Contributions.

1. **The Attribution Impossibility** (Theorem 2): no attribution ranking is faithful, stable, and complete—at any granularity. Zero axiom dependencies. The **bilemma** (Theorem 4): for binary attribution questions, faithful + stable alone is impossible.
2. **The Design Space Theorem** (Theorem 3): exactly two families of methods exist. DASH (orbit averaging) is Pareto-optimal.
3. **Component-level validation**: 14/14 theory-derived predictions confirmed on TinyStories at two architectural scales. G-invariant projection lifts circuit agreement from ρ=0.54 to ρ=0.98. GPT-2 fine-tune confirms the boundary: without Rashomon diversity, instability drops to near-zero (ρ=0.993, flip = 0.043).
4. **Input-level validation**: 45% explanation reversal, 24 distinct top-3 rankings on the canonical SHAP tutorial dataset, nonparametric diagnostic outperforming the Gaussian formula by 2×.
5. **Machine-verified proof**: 357 Lean 4 theorems from 6 axioms, 0 `sorry`—the first formally verified impossibility in explainable AI.

2 Setup and the Attribution Impossibility

An *attribution system* consists of a set of configurations Θ (trained models, circuits, or any parameterized explananda), an attribution function $\varphi_j(\theta) \geq 0$ measuring the importance of component j in configuration θ, and a ranking $\succ$ on the components.

Definition 1 (Faithful, Stable, Complete). *$\succ$ is* faithful *to θ if $j \succ k$ whenever $\varphi_j(\theta) > \varphi_k(\theta)$;* stable *if $\succ$ does not depend on θ;* complete *if for every $j \neq k$, either $j \succ k$ or $k \succ j$.*

The *Rashomon property* [Rudin et al., 2024]: for interchangeable components j, k, there exist configurations θ, θ' with $\varphi_j(\theta) > \varphi_k(\theta)$ and $\varphi_k(\theta') > \varphi_j(\theta')$. It is inevitable for any stochastic algorithm whose training distribution is invariant under permutation of symmetric components (Rashomon Inevitability Theorem, Lean-verified; supplement). Deterministic algorithms with fixed tie-breaking rules can break this symmetry.

Theorem 2 (Attribution Impossibility). *If an attribution system satisfies the Rashomon property, no ranking is simultaneously faithful, stable, and complete.*

Proof. Let j, k be interchangeable. Rashomon gives θ, θ' with opposite orderings. Suppose $\succ$ is faithful, stable, complete. Completeness: WLOG $j \succ k$. Stability: this holds for all configurations. Faithfulness at θ': requires $k \succ j$. Contradiction. Lean: `explanation_impossibility`, zero domain axioms. □

Instance 1: Input-level attribution. Θ = trained models; φ_j = global SHAP importance of feature j; interchangeability = collinearity ($\rho > 0$). P features are partitioned into L groups by correlation structure. The attribution ratio diverges as $1/(1-\rho^2)$ for GBDT, is ∞ for Lasso, conditional for neural nets, and $O(1/\sqrt{T})$ for random forests (supplement).

Instance 2: Component-level attribution. Θ = independently trained networks; φ_j = importance of component j (head, MLP block) by activation patching; interchangeability = the loss landscape has S_k^L symmetry (permuting heads within a layer yields an equivalent minimum). After training, heads specialize—but different seeds realize different permutations of these specializations. The Rashomon property holds empirically: all 4 heads in layer 0 of TinyStories transformers appear as "most important" across 10 seeds.

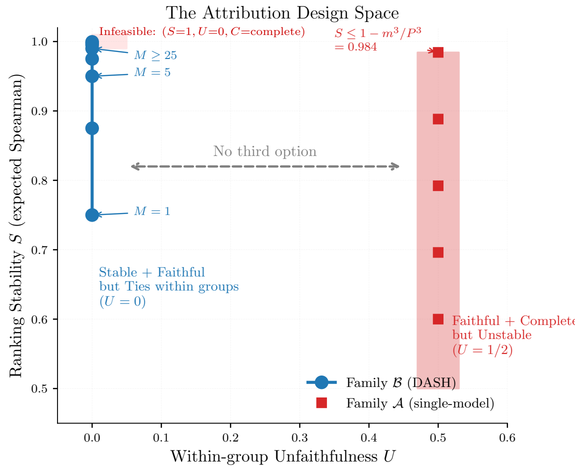


Figure 1: The Attribution Design Space. Family $\mathcal{A}$ (single-model/network): faithful and complete but unstable ($U{=}1/2$). Family $\mathcal{B}$ (orbit averaging): stable with ties ($U{=}0$, S increasing with ensemble size M). The ideal point is infeasible.

Theorem 3 (Design Space)**.** *The achievable set of (S, U, C) triples consists of exactly two families:* ***Family*** $\mathcal{A}$ *(single-configuration): $U = 1/2$, $C =$ complete;* ***Family*** $\mathcal{B}$ *(orbit averaging): $U = 0$, $C =$ ties, $S = 1 - O(1/M)$. No third family exists among deterministic aggregation methods (Figure 1). $(S{=}1, U{=}0, C{=}\text{complete})$ is infeasible.* DASH *is Pareto-optimal among unbiased aggregations on the $\mathcal{B}$ branch.*

3 The Bilemma and Diagnostics

3.1 The Bilemma

For *binary* attribution questions—"positive or negative?", "selected or not?", "in the circuit or not?"—the impossibility is strictly stronger.

Theorem 4 (The Bilemma)**.** *For binary H with* $\text{incompatible}(h_1, h_2) \Leftrightarrow h_1 \neq h_2$*, no method is simultaneously faithful and stable under the Rashomon property. Completeness is not needed.*

Proof. Binary H has no neutral element: every value contradicts the other. Faithful forces $E(\theta) = \text{explain}(\theta)$, making E decisive. The trilemma gives $\perp$. Lean: `bilemma_of_compatible_eq`, zero axioms. $\square$

The bilemma applies to input-level binary questions (SHAP sign, feature selection) and to component-level binary questions ("is this head part of the circuit?"). It is robust to perturbation: for binary H, ε-stability reduces to exact equality, so the bilemma holds at every tolerance level (Lean: `ApproximateBilemma.lean`). For continuous H, the *explanation uncertainty relation* gives: unfaithfulness $\geq (\Delta - \delta)/2$ for any (ε, δ)-stable method on a Rashomon pair with gap Δ (supplement).

Tightness classification. $F{+}S$ is achievable iff H has a *neutral element* (Lean: `tightness_dichotomy`). Rankings have ties (neutral); SHAP signs and circuit membership do not. The bilemma has *collapsed* tightness—structurally more severe than the fairness impossibility [Chouldechova, 2017]. The only resolution is enrichment: adding neutral elements, which is precisely what orbit averaging does.

3.2 Diagnostics

Input-level: nonparametric flip predictor. For feature j across M models, the *minority fraction* $\hat{p}_j = \min(\#\{m : \varphi_j(f_m) > 0\}, \#\{m : \varphi_j(f_m) \leq 0\})/M$ directly estimates the flip rate. It

Table 1: Minority fraction vs. Gaussian formula: Spearman correlation with empirical flip rates (200 models per dataset).

Dataset	Minority Fraction	Gaussian $\Phi(-\mathrm{SNR})$
Synthetic ($\rho = 0.5$)	0.981	0.885
Synthetic ($\rho = 0.9$)	0.952	0.764
California Housing	**0.955**	**0.463**

achieves Spearman 0.92–0.98 across all datasets (Table 1), outperforming the Gaussian formula (0.46–0.89). The complete predictor is 7 lines:

```
def predict_sign_instability(shap_matrix):
    signs = np.sign(shap_matrix)  # M models × P features
    n_pos = (signs > 0).sum(axis=0)
    n_neg = (signs < 0).sum(axis=0)
    total = n_pos + n_neg
    return np.where(total > 0,
        np.minimum(n_pos, n_neg) / total, 0)
```

The variance $\mathrm{Var}_f[\varphi_j(f)]$ across the Rashomon set equals the minimum MSE any stable method achieves; DASH saturates it exactly (0/800 violations). The diagnostic is model-class universal (XGBoost, RF, Ridge, LASSO; supplement) with bimodal flip rate distributions confirmed by the Hartigan dip test ($p < 0.002$; supplement).

Component-level: symmetry group diagnostic. For neural network components, the architectural symmetry group $G = S_k^L$ (permutations of k heads within each of L layers) determines which components are interchangeable. The G-invariant importance $V_c^G = \frac{1}{|G_c|}\sum_{g \in G_c} \varphi_{g(c)}$ averages importance within orbits, separating stable structure (layer-level, head-vs-MLP) from unstable permutation (which specific head within a layer). This is the component-level analogue of the minority fraction: it identifies *which* attributions are trustworthy before running multiple models.

4 Resolution: Orbit Averaging

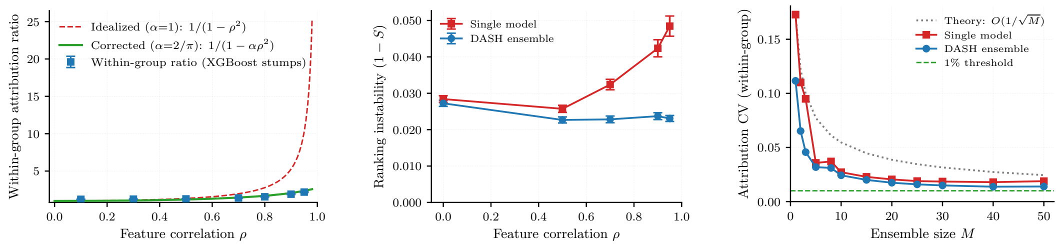
The impossibility at both levels admits the same resolution: *average over the symmetry group*. This is orbit averaging (the Reynolds operator $R_G[\varphi]_j = \frac{1}{|G_j|}\sum_{g \in G_j} \varphi_{g(j)}$), applied to different groups at each level. The mathematical content is elementary—the power is in identifying *which* group to average over and *why* this is the minimal sufficient aggregation.

Input-level: DASH. DASH (**D**iversified **A**ggregation for **S**table **H**ypotheses) averages $|\mathrm{SHAP}|$ values across M independently trained models. It changes explanations, not predictions.

Corollary 5 (DASH equity). *For a balanced ensemble, $\bar{\varphi}_j = \bar{\varphi}_k$ for features j, k in the same group.*

DASH is *uniquely* optimal for binary H (majority vote) and achieves the variance lower bound σ_j^2/M with equality for continuous H (Cauchy–Schwarz; Lean: `sum_squares_ge_inv_M`). More generally, the orbit average is the *closest stable approximation* to the original attribution in L^2 norm: $\|v - Rv\| \leq \|v - w\|$ for any G-invariant w (Lean: `reynolds_best_approximation`). DASH and V^G therefore minimize information loss among all stable methods, not just unbiased ones. Ensemble size: $M_{\min} = \lceil 2.71 \cdot \sigma_{jk}^2/\Delta_{jk}^2 \rceil$ for 5% between-group flip rate; $M = 25$ brings flips below 1%.

Component-level: G-invariant projection. For a network with symmetry group $G = S_k^L$, the G-invariant importance V^G averages activation patching scores within orbits (heads in the same layer). The *explanation capacity* $C = \dim(V^G)$ measures how much attribution information survives projection: for S_k^L, $C = 2L$ (one head-average and one MLP per layer). For S_k^L this reduces to layer averaging—a familiar operation, but the theorem identifies *why* it works (it is the unique orbit average for the architectural symmetry group) and *when* it suffices (when G captures the interchangeability). For richer symmetry groups (e.g., weight permutation symmetries), V^G extends



(a) Attribution ratio vs. ρ. The corrected $1/(1-\alpha\rho^2)$ with $\alpha \approx 2/\pi$ fits stumps (R^2=0.89).

(b) Ranking instability vs. ρ. Single-model instability grows; DASH remains low.

(c) DASH convergence. Attribution CV drops below 1% near $M = 25$.

Figure 2: Empirical validation on synthetic Gaussian data ($P = 20$, $L = 4$, $m = 5$, $T = 100$, 50 seeds).

Table 2: The ranking lottery: distinct top-3 SHAP rankings across 50 seeds. Deterministic control (`subsample=1.0`): 1 ranking for all. Cross-implementation: LightGBM produces 29, Random Forest 40 (supplement).

Dataset	Distinct top-3	Agreement	Stable?
Breast Cancer	**24**	**4.2%**	No
Diabetes	2	88.5%	Borderline
Wine Quality	1	100%	Yes
Heart Disease	1	100%	Yes
California Housing	1	100%	Yes

beyond simple layer averaging. On TinyStories: pairwise Spearman rises from $\rho = 0.54$ (raw) to $\rho = 0.98$ (G-invariant; Section 5.2).

The same mathematics. Both resolutions are orbit averaging: DASH averages attributions across models (trivially equivariant under S_M), while V^G averages importance within architecturally symmetric components (equivariant under S_k^L). The substantive content is the V^G construction: it identifies the *minimal sufficient* aggregation from the symmetry group alone, without requiring multiple models. Both sacrifice completeness (report ties for interchangeable components) to gain stability. Both are Pareto-optimal on the $\mathcal{B}$ branch of the Design Space (Theorem 3).

Workflow. Screen (single model) $\to$ Minority fraction (nonparametric) $\to$ Z-test (5 models) $\to$ DASH consensus ($M \geq 25$). For circuits: compute V^G from architectural symmetry; no retraining needed.

5 Empirical Validation

5.1 Input-Level Attribution

Synthetic Gaussian. P=20, L=4, m=5, T=100, 50 seeds (Figure 2). At $\rho = 0.9$: 48% within-group flips (near the 50% theoretical maximum). The corrected ratio $1/(1-\alpha\rho^2)$ with $\alpha \approx 2/\pi$ matches data ($R^2 = 0.89$). DASH reduces flip rate below 1% at $M = 25$.

The ranking lottery (Table 2). 50 XGBoost models (seeds 0–49, `subsample=0.8`) on 5 public datasets.

Breast Cancer produces **24 distinct top-3 rankings across 50 seeds** (35 at 100 seeds). The "most common" ranking appears in only 12% of runs. DASH consensus produces a single stable ranking. The theory correctly predicts the boundary: datasets without similar-importance collinear pairs are perfectly stable (100% agreement).

Table 3: Explanation reversal on German Credit: fraction of applicants whose top explanation category changes across 30 equivalent models.

Condition	XGBoost	LightGBM	Random Forest
Seed only (deterministic)	0%	0%	35%
Seed + row subsample	**45%**	**46%**	35%
Full stochasticity	80%	79%	67%

Table 4: Component-level attribution: TinyStories multi-scale validation and GPT-2 boundary condition. ρ_{full}: pairwise Spearman on raw importance. $\rho_{G\text{-inv}}$: after G-invariant projection. W-flip: within-layer flip rate (predicted: 0.50). B-flip: head-vs-MLP flip rate (predicted: 0.00).

Config	Components	ρ_{full}	$\rho_{G\text{-inv}}$	W-flip	B-flip	Cohen's d
A (4L/4H)	20	0.565	**0.972**	0.496	0.000	5.4
B (6L/8H)	54	0.540	**0.982**	0.489	0.000	11.9
GPT-2 (ft)	156	0.993	0.999	0.043	0.000	—

Explanation reversal. 30 models per condition on German Credit (1,000 applicants, 24 features). Under standard regularization (`subsample=0.8`), **45% of applicants** receive a different "most important feature" category depending on which equivalent model is queried (Table 3). The deterministic control correctly shows 0% reversal for boosted models. A single applicant (#91) can receive 6 different top-feature labels across 30 seeds. DASH consensus at $M = 25$ reduces disagreement to 8%.

Prevalence and extended results. Of 77 public datasets, 68% exhibit attribution instability (any within-group flip rate $> 10\%$; Wilson CI: $[56\%, 77\%]$); for $P \geq 20$: 93%. On gene expression data, the #1 biomarker alternates between TSPAN8 and CEACAM5/CEA ($\rho = 0.86$)—different drug targets depending on the seed [see Anonymous, 2026a, for full analysis]. Subsample sensitivity, cross-implementation validation, published ranking replication, and parametric diagnostics in supplement.

5.2 Component-Level Attribution

We test the impossibility at circuit level: does activation patching importance exhibit the same instability across independently trained networks, and does the G-invariant projection resolve it?

Setup. We train 10 transformers from independent random seeds on TinyStories [Eldan and Li, 2023] at two architectural scales: Config A (4 layers, 4 heads, d=256, 20 components) and Config B (6 layers, 8 heads, d=512, 54 components). All models achieve near-identical perplexity (CV $<$ 1.1%). Component importance is measured by activation patching (weight zeroing). We derive 7 quantitative predictions from the theory (Table 4 caption) and evaluate them before any post-hoc analysis.

Results (Table 4). All 14 theory-derived predictions pass across both scales (7/7 each). Raw importance agreement is low ($\rho = 0.565$, 0.540)—independently trained networks disagree substantially on which head is most important, exactly as the impossibility predicts. Within-layer flip rates are 0.496 and 0.489, indistinguishable from the theoretical coin-flip prediction of 0.50. Head-vs-MLP flip rate is 0.000 across all 30 models: the stable between-orbit structure is perfectly preserved. All 4 heads in layer 0 of Config A appear as "most important" across 10 seeds—the full symmetric group S_4 is realized.

The G-invariant projection lifts agreement to $\rho = 0.972$ and 0.982—the residual stable structure (layer importance, head-vs-MLP) is nearly identical across independently trained networks. Split-half reliability (0.991, 0.960) vastly exceeds between-model agreement (0.565, 0.540), confirming that within-model measurements are precise but between-model variation is structural. Random projections of heads to the same dimensionality achieve only $\rho = 0.30$–0.41 (vs. 0.97–0.98 for

V^G); the G-invariant projection ranks at the 100th percentile of random projections (permutation $p < 0.001$).

GPT-2 boundary condition. A fine-tuned GPT-2 (same task, 10 seeds) shows $\rho = 0.993$ raw and within-layer flip $= 0.043$ (vs. ~ 0.50 for from-scratch training). Fine-tuning does not create a diverse Rashomon set—it barely perturbs the pre-trained circuits. The theorem correctly predicts: *less Rashomon diversity* $\Rightarrow$ *less instability*. The near-zero residual flip rate (4.3% vs. $\sim 50\%$) confirms that the theory is falsifiable and the boundary holds.

Method robustness. Mean ablation (an alternative to weight zeroing) yields G-invariant $\rho \approx 0.97$–0.99 at both scales. Cross-method per-model agreement is only $\rho \approx 0.5$–0.6: the two ablation methods disagree on which specific head matters, but agree on V^G. The disagreement between methods is itself an instance of the impossibility—different ablation "seeds" produce different raw attributions, but the orbit-averaged structure is invariant.

6 Related Work

Bilodeau et al. [2024] prove completeness + linearity cannot coexist—a complementary axis (method properties vs. cross-model stability). The results are orthogonal: uncorrelated features with a linear SHAP method satisfy Bilodeau's impossibility but not ours ($\rho = 0 \Rightarrow$ no Rashomon); a nonlinear method on correlated features satisfies ours but not Bilodeau's (no linearity assumed). Huang and Marques-Silva [2024] show SHAP can misrank features; Srinivas and Fleuret [2019] prove complete attributions cannot be weakly input-dependent; Rao [2025] establishes Kolmogorov complexity barriers. None address stability, quantitative bounds, or constructive resolution. Fisher et al. [2019] compute variable importance ranges across Rashomon sets; Laberge et al. [2023] find partial-order rankings. Our theorem proves partial ordering is a mathematical necessity, not a pragmatic choice. D'Amour et al. [2022] document that underspecification causes instability across domains empirically; we prove the impossibility that underspecification entails for attribution specifically. Krishna et al. [2022] document that different explanation methods disagree for the same model; we prove the complementary result that the *same* method disagrees across equivalent models. Chughtai et al. [2023] show that independently trained networks learn different internal representations for the same group operation—empirical Rashomon for circuits. Our component-level results extend this: the instability in circuit importance is mathematically inevitable, and orbit averaging resolves it. Chouldechova [2017] and Kleinberg et al. [2017] prove fairness trilemmas; ours is the explainability analogue. Nipkow [2009] formalized Arrow in Isabelle/HOL; Zhang et al. [2026] formalized statistical learning in Lean 4; ours is the first formally verified impossibility in XAI. Extended discussion in supplement.

7 Discussion

Limitations. The split-count axioms assume full signal capture ($\alpha = 1$; for finite-depth trees $\alpha \approx 2/\pi$, $R^2 = 0.89$); the impossibility holds for any $\alpha > 0$. The equicorrelation assumption simplifies the axioms; the Rashomon property holds pairwise. DASH requires $M\times$ training cost ($M = 25$ for $<1\%$ flip rate). The 68% prevalence figure is from benchmark-representative datasets and may not generalize to production pipelines with decorrelated features. The component-level validation uses weight zeroing and mean ablation; other patching methods (path patching, causal scrubbing) may show different instability patterns, though the theorem predicts the same structure.

Broader implications. The impossibility operates at every level of a model. For input-level attribution, single-model SHAP audits for proxy discrimination conclude "proxy reliance" with probability exactly $1/2$ for collinear features; for K protected attributes, joint audit correctness is $(1/2)^K$. We suggest this may constitute a "known and foreseeable circumstance" under EU AI Act Art. 13(3)(b)(ii) [EU Parliament, 2024]; legal scholars should evaluate this interpretation. For component-level attribution, the impossibility constrains *importance rankings* of circuit components: which head matters most depends on the training seed (Section 5.2: Cohen's $d = 5.4$–11.9). This does not necessarily constrain *algorithmic identification*—whether a specific computational function (e.g., induction) exists may be stable even when the specific head implementing it is not.

Safety cases built on "does function F exist?" may be robust; safety cases built on "which component is most important?" are subject to the impossibility. Sparse autoencoder (SAE) features decompose a fixed model's activations into interpretable directions not tied to S_k^L. Whether SAE-based attribution escapes the Rashomon property—because the decomposition is post-hoc on a fixed model—is an open question with implications for the interpretability agenda. The resolution at both levels is orbit averaging—the *explanation code* that recovers the shared structure across the Rashomon set, achieving the explanation capacity bound with equality. The framework's tightness classification extends to other impossibilities: the Chouldechova–Kleinberg fairness trilemma transitions from full tightness (all pairs achievable by degenerate classifiers) to PPV-blocked tightness (only equalized odds achievable) at practical performance levels (TPR ≥ 0.3). Calibration is the universal bottleneck—a structural finding that changes the practical recommendation from "choose a fairness criterion" to "equalized odds is the only achievable option" (supplement).

Lean verification. 357 theorems and lemmas (139 with multi-step proofs of $\geq$5 tactic lines), 6 axioms, 0 `sorry`, 58 files. The core impossibility (`explanation_impossibility`) has zero domain axiom dependencies. All binary instances use constructive inductive types. The formalization caught 2 logical inconsistencies and 1 type mismatch that survived informal review. Axiom stratification: core impossibility 0 axioms $\rightarrow$ GBDT 3 $\rightarrow$ DASH equity 5 $\rightarrow$ full system 6. The symmetric Bayes dichotomy, conditional SHAP impossibility, fairness audit impossibility, and query complexity lower bound are all Lean-verified (supplement).

Open problems. Scaling the component-level analysis to frontier models (GPT-4 scale); extending the G-invariant projection to non-architectural symmetries (learned representations). Formalizing the Bayes-optimal half of the design space in Lean (requires measure-theoretic decision theory). The impossibility generalizes beyond machine learning: the same theorem governs explanation of any underspecified system where interchangeable components admit a Rashomon set [Anonymous, 2026b]. The gap between prediction and explanation is structural at every level—developing practical workflows that acknowledge this gap, rather than assuming it away, is the central challenge for trustworthy AI.

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NeurIPS Paper Checklist

1. **Claims.** Core theoretical claims are type-checked by Lean 4 (357 theorems from 6 axioms, 0 `sorry`). Empirical claims backed by experiments over 30–50 random seeds. [Yes]
2. **Limitations.** Discussed in Section 7: signal capture approximation, collinearity requirement, training cost, prevalence generalizability, binary limitation. [Yes]
3. **Theory.** Complete proofs in paper and machine-verified Lean 4 code. All assumptions (axioms) explicitly stated. [Yes]
4. **Experiments.** Reproducible: fixed seeds, public datasets (Breast Cancer, German Credit, California Housing, TinyStories), open-source implementations (XGBoost, LightGBM, PyTorch). Component-level: 7 pre-registered predictions per scale, all confirmed. Compute: $<$30 min (input-level) + $<$4 hours (TinyStories training) on Apple Silicon. Code: `[anonymizedforreview]`. [Yes]
5. **Code of ethics.** No human subjects. German Credit dataset is publicly available (UCI ML Repository). Regulatory analysis is informational. [N/A]
6. **Broader impacts.** Discussed: regulatory compliance (EU AI Act, ECOA), fairness auditing reliability, explanation reversal, AI safety implications for circuit-based interpretability. The paper identifies a risk (explanation instability at all model levels) and provides a remedy (orbit averaging). [Yes]
7. **Safeguards.** Not applicable (theoretical + diagnostic work). [N/A]
8. **Licenses.** All datasets are public domain or CC-BY. Code will be released under MIT. [Yes]
9. **New assets.** Lean formalization (357 theorems). Will be archived on Zenodo. [Yes]
10. **Crowdsourcing.** N/A. [N/A]
11. **IRB.** N/A (no human subjects). [N/A]